OPENCAMPUS.SH · SESSION 2

Data import and visualization

On which weekdays should we bake more, and how sure can we be?

TODAY

Plan for the session

1. Quiz on last week’s videos
2. Live demo: set up and grill a question
3. Your week 1 answers in groups
4. Working with the agent; scale levels and charts
5. Group work: on which weekdays should we bake more?
6. Break
7. Show your results, with feedback
8. Homework and what comes next

QUIZ · PLENARY · 10 MINUTES

Five questions on last week, two on how it went



forms.cloud.microsoft/e/sgF9UkxXk2

About 4 minutes to answer, then we discuss. Not scored: we look at how the answers spread and talk about the questions most of you got wrong.

LIVE DEMO · 10 MINUTES

From an empty repository to a sharp question

1 · Set up

```
/setup-ds-project
```

The agent asks a few questions, each with a recommendation. Afterwards the folder has CANVAS.md, GLOSSARY.md, DECISIONS.md and checks/.

2 · Grill the question

```
/grill-question On which weekdays should we bake more, and how sure can we be?
```

It asks one question at a time and waits. You decide; it writes blocks 1–4 of the canvas.

Watch for: where does the agent stop and wait for a decision?

INPUT · HOW SKILLS WORK

Some skills you start, some the agent loads, and some wait for you

Commands you type

You start them. In Codex: $ instead of /.

`/setup-ds-project`

`/grill-question`

Reference skills

The agent loads them when they fit. This week, when it writes for the operations manager:

`audience`

Stop points

The agent waits for your decision, as in /grill-question. Facts are its job; decisions are yours.

HOMEWORK REVIEW · GROUPS OF THREE · 10 MINUTES

Which answer would the operations manager act on?

Form	Groups of three. You stay together for the group work today.
Read	Each of you reads out your answer on the seasonal bread from week 1, the last Markdown cell of your notebook.
Pick	Agree on the one the manager would act on: **correct**, **understandable** without explanation, **actionable**. What would make it better?

Then two groups read their pick to the room and say why: 3 minutes.

INPUT · WORKING WITH THE AGENT

Give the agent context, then a plan, then questions

Context files

AGENTS.md is the entry point: data, conventions, and where to find **CANVAS.md** (the decision), **DECISIONS.md** (what you decided, and why) and **GLOSSARY.md** (your terms).

Plan first

Ask for a plan before any code. Read it, change it, then let the agent run.

Ask for explanations

What does each step do, and why this way? If you cannot follow it, you cannot check it.

INPUT · WORKING AS A TEAM

Decide together at the stop points

Alone

The agent’s proposal sounds reasonable, so you accept it.

Together

Someone asks “why?”, someone else sees what is missing, and you decide.

In your group today: the driver types, the other two read every proposal and ask why before you accept it.

INPUT · SCALE LEVELS

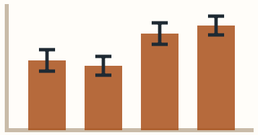
The scale level decides what you may compute and say

Level	Bakery example	What makes sense
Nominal	Product group	Counts and shares; the order of the bars is free
Ordinal	Weekday	Order matters: Monday to Sunday, never alphabetically
Interval	Temperature in °C	Differences, but 20 °C is not “twice as warm” as 10 °C
Ratio	Turnover	Mean, confidence interval and “twice as much”

Today: the weekday sets the order of your bars; the turnover is what you average.

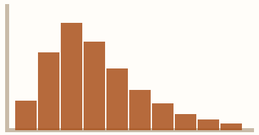
INPUT · CHART CHOICE

Pick the chart from the question



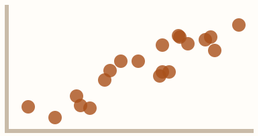
Compare averages across groups

Bar chart with confidence intervals



Show how one number is distributed

Histogram or box plot



Relate two numbers

Scatter plot



Follow a number over time

Line chart



Show the shares of one whole

Pie chart, with few slices



Compare a number across regions

Choropleth map

INPUT · THE CHECKLIST

Check five things before you trust a chart

1. Does the number of rows match the CSV, and are the values plausible?
2. Are the weekdays in order, Monday to Sunday, and how many days stand behind each bar?
3. What do the error bars show?
4. Does the pattern hold in every product group?
5. Does the chart answer the operations manager’s question?

Ask the agent, then look yourself: today before you show your results, and again at home.

GROUP WORK · 30 MINUTES · SAME GROUPS

On which weekdays should we bake more, and how sure can we be?

Who	One driver at the keyboard; rotate every 10 minutes, so everyone drives once.
Where	The driver’s practice repository, notebook **week-02-weekdays.ipynb** in its main folder.
Data	umsatzdaten_gekuerzt.csv, from the URL in AGENTS.md.
Task	Average daily turnover (all products added up per day) for each weekday, as a bar chart with 95 % confidence intervals, Monday to Sunday. Give it an action title that states the conclusion.
Done when	One chart with seven bars, error bars and a title, checked with the five points, ready to show.

OPTIONAL · IF YOUR GROUP FINISHES EARLY

Does it differ by product group?

Extend	The same chart for the six product groups, one subplot per group: the groups sell very different amounts.
Compare	Which groups follow the weekday pattern, and which do not?
Recommend	Draft a recommendation for the operations manager: what should production planning do per weekday?

5 MINUTES

Break

SHOW YOUR RESULTS · PLENARY · 17 MINUTES

Would the operations manager act on your chart?

1

Does the title answer the question?

2

Would the manager understand it without explanation?

3

Are the confidence intervals read correctly?

PLENARY · 3 MINUTES

What was hard, and what surprised you?

Where did you get stuck?

What did the agent do that you did not expect?

What will you do differently at home?

HOMEWORK · ABOUT 3 HOURS · ALL DETAILS IN THE COURSE HANDBOOK, WEEK 2

Until next session

1 · Finish the weekday analysis	Watch the six videos first (≈ 22 min), then build your own week-02-weekdays.ipynb; the handbook has a starter prompt.
2 · Check and clean up	The five points from today, then Restart and Run All without errors.
3 · Write the memo	Update the course skills, draft it with the prompt from the handbook, then rewrite the title and the recommendation yourself.
4 · Watch the other videos	Questions in the Element course room.

DELIVERABLE · FOR THE OPERATIONS MANAGER

A memo of half a page

An action title

It states the conclusion, not the topic.

A recommendation

For production planning per weekday.

How sure we can be

In plain words, from your intervals.

One chart

The one that supports the recommendation.

Save it as week-02-memo.md in your practice repository and bring it to the next session.

NEXT WEEK · SESSION 3

Your team and your project

Teams of three or four start a fresh repository from the template and take on the project question together, with Git.